

Hiranandani Fortune City — "Arena" Web App

Full QA & Technical Audit Report · Repository `C:\Users\FS165\hfc` (branch `master`, commit `7beebdb`)
Date: 8 September 2026 · Stack: React 19.2 + TypeScript 5.9 + Vite 7.1 + Tailwind 3.4 + MUI 7 + React Router 7

1. Executive Summary

The application was tested end-to-end: dependency install, TypeScript compile, ESLint, production build, automated headless-Chrome walkthrough of **all 29 static routes plus 29 dynamic floor/unit routes**, data-integrity validation of all 8 tower datasets (322 floors / 1,747 units), external media reachability, mobile-viewport layout, page-weight/performance measurement on the production build, and a security audit.

Overall the app is functional — it builds cleanly, TypeScript reports zero errors, every route renders, all 11 external video/panorama embeds are live, and the tower data has zero integrity defects. However testing found **2 reproducible crashes that blank the entire screen, 2 broken panorama pages**, and severe page-weight problems (one page downloads 43 MB). These must be fixed before the next release; the remainder are quality, accessibility and maintenance issues.

58	2	4	17	0	67
ROUTES TESTED	CRITICAL CRASHES	HIGH PRIORITY	MEDIUM / LOW	TS ERRORS	ESLINT ERRORS

Test results at a glance

Check	Result	Detail
Dependency install (<code>npm install</code>)	FAIL	ERESOLVE peer conflict; needs <code>--legacy-peer-deps</code>
TypeScript compile (<code>tsc -b</code>)	PASS	0 errors
Production build (<code>npm run build</code>)	PASS	Built in 22.6 s; 1 chunk-size warning
ESLint	FAIL	67 errors, 1 warning (incl. 2 React hook-order violations)
Route rendering — 29 static routes	PASS	All render; 404 page works for unknown paths
Route rendering — floor/unit routes (8 towers)	PASS	Valid + invalid IDs both handled
Tower data integrity (8 datasets)	PASS	322 floors, 1,747 units, 0 duplicate/missing IDs or images
Asset import resolution (587 references)	PASS	0 broken imports; 0 broken <code></code> at runtime
External media embeds (11 URLs)	PASS	Wistia, Vimeo & Pano2VR tours all load and play
Internal panorama embeds (2)	FAIL	<code>/pano/...</code> folder missing → 404 rendered inside iframe
Interactive controls (gallery, slider, hotspots)	FAIL	Gallery "Previous" button crashes the app
Mobile layout 390×844 (6 pages)	PASS	No horizontal overflow on any page tested

Page weight / performance	FAIL	Up to 43.5 MB per page load
Security audit (<code>npm audit</code>)	FAIL	18 vulnerabilities (13 high) — all build-time tooling
Accessibility basics	FAIL	52 of 61 images have no <code>alt</code> text

2. Critical Findings (P0) — fix before release

P0-1 Gallery "Previous" button crashes the whole application

File: `src/Pages/Arena/GalleryPage.tsx:310` · **Route:** `/arena_gallery` · **Status:** reproduced automatically

What happens: On the gallery page, clicking the "◀ previous" arrow while the *first* image is showing turns the entire screen white. The React root is emptied (`rootChildren = 0`) and only a full browser reload recovers it. On a sales-lounge kiosk this looks like a total app failure.

Console error captured: `TypeError: Cannot read properties of undefined (reading 'src')`

Cause: the handler uses a plain modulo, which returns `-1` at index 0:

```
onClick={() => setCurrentIndex((prev) => (prev - 1) % images.length)}
```

`images[-1]` is `undefined`, and the very next line reads `images[currentIndex].title`, throwing during render.

Fix: wrap the index the way the lightbox handler already does elsewhere in the same file:

```
setCurrentIndex((prev) => (prev - 1 + images.length) % images.length)
```

P0-2 React hook-order crash on Pavilion & Olympus unit pages

Files: `src/Towers/Tower3_Pavilion/Unit_Pavilion.tsx:49`, `src/Towers/Olympus/Unit_Olympus.tsx:50` · **Status:** reproduced automatically

What happens: both files call `useEffect` *after* an early `return` for the "Unit not found" state. When the page moves from a valid unit to a missing unit without remounting (browser back/forward between unit URLs, or any in-app navigation that changes only the `:id`), React sees fewer hooks than the previous render and unmounts the whole tree — white screen again.

Console error captured: `Rendered fewer hooks than expected. This may be caused by an accidental early return statement.`

Also flagged statically by ESLint: `react-hooks/rules-of-hooks` (2 errors).

Fix: move the `useEffect` block above the `if (!singleUnit) return ...` guard. The other six tower unit pages already do this correctly and are unaffected.

3. High Priority (P1)

P1-1 Ebony & Golden Willows entrance panoramas are broken

Files: `src/Pages/Ebony/EntrancePageEbony.tsx:14`, `src/Pages/GoldenWillows/EntrancePageGoldenWillows.tsx:13`

Routes: `/ebony`, `/goldenwillows`

Both pages embed `<iframe src="pano/pano3/index.html"> / "pano/pano1/index.html"`, but no `pano/` folder exists in `public/`. Because the SPA fallback (`public/_redirects` → `/* /index.html 200`) answers every unknown path with the app itself, the iframe loads the app *inside* the app and displays its own **"404 — Oops! The path is mismatched..."** page. Verified in-browser: the iframe's document text is literally the 404 screen. This will behave identically on Netlify in production.

Fix: either ship the Pano2VR tour folders into `public/pano/`, or point the iframes at the hosted tours the way `/` and `/arena` already do (`https://futeservices.com/.../index.html`). Also add a `/pano/*` exclusion to `_redirects` so a genuinely missing tour shows a 404 instead of the app.

P1-2 Extreme page weight — up to 43.5 MB for a single page

Measured on the production build (`vite preview`, cache disabled, 1440×900)

Route	Transfer	Images	Largest single asset
<code>/club-house</code>	43.47 MB	42.99 MB	club4.webp — 10.81 MB
<code>/arena_gallery</code>	27.94 MB	27.46 MB	interior13.webp — 2.92 MB
<code>/arena_stadia</code>	7.47 MB	6.99 MB	stadia_updated.png — 6.99 MB
<code>/arena_master-plan</code>	6.83 MB	—	MASTERPLAN-55.png — 6.35 MB
<code>/arena_projecthighlights</code>	3.61 MB	2.83 MB	exteriortwr7.webp — 2.83 MB
<code>/</code> (entrance)	0.67 MB	0.19 MB	index.js — 0.33 MB (gzip)

On a typical 4G connection (~5 Mbps) the club-house page needs roughly **70 seconds** to finish loading. Contributing factors:

- **Gallery thumbnails use the full-resolution source.** The 96×96 px thumbnail strip loads the same multi-megabyte originals as the main viewer — no separate thumbnail asset exists.
- **Uncompressed masterplan/tower PNGs.** `exterior5.png` alone is 34 MB in `dist/`; six more assets exceed 6 MB. Converting the large PNGs to WebP/AVIF and capping the longest edge at ~2560 px would cut most pages by 70–90% with no visible quality loss.
- **The full build output is 561 MB** across 337 asset files.

P1-3 A clean `npm install` fails

On a fresh checkout `npm install` aborts with `ERESOLVE : react-18-image-lightbox@5.1.4 declares peer react@"16.x || 17.x || 18.x"` but the project runs React 19. Any new developer or CI pipeline is blocked until they discover `--legacy-peer-deps`.

Fix (pick one): replace `react-18-image-lightbox` with a React-19-compatible lightbox (e.g. `yet-another-react-lightbox`); or commit an `.npmrc` containing `legacy-peer-deps=true` so installs are reproducible; or add a package `overrides` entry. The library does work at runtime today — this is an install-time blocker, not a functional one.

P1-4 18 npm security advisories (13 high)

Affected: `vite` (6 advisories — dev-server path traversal / arbitrary file read), `rollup`, `@babel/core`, `brace-expansion`, `browserslist`, `ajv`, `yaml`, `@humanfs/node`, `launch-editor`.

All are **build-time tooling only** — none ship in the browser bundle, so the deployed site is not directly exposed. The Vite dev-server advisories do matter for developer machines. `npm audit fix` resolves the whole set; every runtime dependency is also 1–2 minor versions behind (React 19.2.0 → 19.2.8, React Router 7.9.3 → 7.18.3, MUI 7.3.4 → 7.3.11).

4. Medium Priority (P2) — correctness & UX

P2-1 Duplicate React keys — master-plan hotspots can silently disappear

Three datasets reuse the same `id` across different entries, and that `id` is used as the React `key`. React warns that children "may be duplicated and/or omitted". Captured live on three routes.

File	Duplicate IDs	Effect
<code>src/data/Aminities.ts</code>	4, 14, 21, 22, 23, 54	e.g. three separate "Drop Off node" hotspots all share <code>id: 4</code> at different coordinates — on <code>/arena_master-plan</code>
<code>src/data/GoldenMasterPlanData.ts</code>	1, 3, 4, 20, 21, 29, 43, 46, 47	on <code>/goldenwillowslayout</code>
<code>src/Pages/Club_House/ClubHouse.tsx:64</code>	3 (used twice in <code>data1</code>)	on <code>/club-house</code>

Fix: the numeric `id` is the on-screen label and is legitimately repeated, so keep it for display but key the elements on the array index (or a composite ``${id}-${coords}``).

P2-2 Missing key on the Olympus tooltip list

`src/Towers/Olympus/Olympus.tsx:145` maps 41 floors to `<Tooltip>` elements without a `key`, producing "Each child in a list should have a unique key prop" on `/arena_olympus`. `Stadia.tsx:66` shows the correct pattern (`key={`tooltip-${floor.id}`}`); the other six towers render a single tooltip and are unaffected.

P2-3 Master-plan amenities legend is invisible at every screen size (typo)

`src/components/MasterPlanDescription.tsx:5` uses `className='... hidden lg:bloack'`. `lg:bloack` is a typo for `lg:block`, so Tailwind generates no rule and the `hidden` class is never overridden. The whole "Tower-wise indoor amenities / Proposed indoor amenities" panel therefore never renders — on desktop or anywhere else.

P2-4 Navigation bar disappears on two of its own menu destinations

`App.tsx:66` lists the routes that show `<Navbar />`, but omits `/arena_gallery` and `/project_status` — both of which *are* items in that navbar. Verified: those two pages render 0 nav links, every other Arena page renders 7–8. Users must use the small circular back button to escape.

The Golden Willows navbar (`App.tsx:71`) has the same gap for `/goldenwillowslayout` and `/gallery_golden`.

P2-5 Gallery play/pause icon is inverted

`GalleryPage.tsx:324` renders `{paused ? <MdPause /> : <MdPlayArrow />}` — it shows a "play" icon while the slideshow is playing and a "pause" icon while it is paused. `PureSlider.tsx:91` implements the same control the conventional way round. One of the two is wrong; the gallery is the odd one out.

P2-6 Opening a unit needs a double-click in 7 towers, a single click in 1

Arcadia navigates to the unit page on `onClick` (`Floor_Arcadia.tsx:131`); Pavilion, Grandstand, Citadel, Greenfield, Athletica, Stadia and Olympus all require `onDoubleClick` . On a touchscreen sales kiosk a double-tap is unreliable and undiscoverable — visitors are likely to conclude the unit plans don't open at all. Recommend standardising on single click/tap everywhere.

P2-7 Debug logging left in production

`console.log("singleFloor", singleFloor)` runs on every render of 7 floor pages: `Floor_Arcadia:22` , `Floor_Citadel:22` , `Floor_Grandstand:22` , `Floor_Greenfield:22` , `Floor_olympus:24` , `Floor_Stadia:24` , `Floor_Pavilion:24` . Each logs a large object containing every unit and room on the floor.

P2-8 Favicon is a 157 KB PNG served as `image/svg+xml`

`index.html:5` declares `<link rel="icon" type="image/svg+xml" href="/vite.svg">` , but `public/vite.svg` is not an SVG — it is a byte-identical copy of `logo2.png` (verified by checksum). The declared MIME type doesn't match the content, so browsers may refuse to display it, and it costs 157 KB. Ship a real `favicon.ico` / `.png` at 32–64 px and correct the `type` .

P2-9 Accessibility & SEO basics missing

- **52 of 61** `<img>` tags have no `alt` attribute — including both header logos on every page and all gallery/club-house images.
- `index.html` has no `meta description` and no Open Graph / Twitter card tags, so links shared on WhatsApp or social media show no preview — a real cost for a property marketing site.
- Six `<a>` elements are nested inside `<button>` on the entrance page (`EntrancePage.tsx`), which is invalid HTML and confuses keyboard and screen-reader navigation. Style the `<Link>` itself instead.
- No page defines a document `<h1>` on most routes; the `<title>` is the same "Hiranandani Communities" for all 58 routes.

P2-10 67 ESLint errors

Beyond the 2 hook-order errors (P0-2), the bulk are: **40** × `no-explicit-any` , **14** × `no-unused-vars` (dead imports and unused state such as `isLargeScreen` in `Stadia.tsx:14` and `setSelectedUnit` in `Floor_olympus.tsx:20`), **9** × `no-wrapper-object-types` (`Number` used instead of `number`), plus `prefer-const` and `ban-ts-comment` . `npx eslint . --fix` clears 9 automatically.

5. Low Priority (P3) — maintenance & housekeeping

Area	Finding
Repository bloat	<code>src/assets/</code> is 904 MB , of which 218 MB across 118 files is never referenced by any code — abandoned master-plan revisions (<code>arena_masterplan1-4</code> , <code>MASTERPLAN-5.png</code> , <code>masterplan_new_18_7_2026.webp</code>), an unused <code>gallery1/</code> folder, and superseded tower renders. The git pack is 772 MB, making clones very slow. Consider deleting the unused files and moving large binaries to Git LFS or a CDN.
Dead code	<code>src/Pages/Arena/GalleryPage1.tsx</code> (321 lines) is imported nowhere; <code>src/Pages/GoldenWillows/Towers_Golden/{MainTowerPage,FloorPage,UnitPage}.tsx</code> (682 lines) are unused with their routes commented out in <code>App.tsx:145-147</code> . Four files carry 230–300 commented-out lines each.
Unreachable routes	<code>/arena_quality</code> is registered but its navbar entry is commented out (<code>Navbar.tsx:13</code>) — no link reaches it. <code>/ebony</code> is registered but the entrance page links to the external <code>ebony.futeservices.in</code> instead, so the (broken, see P1-1) internal page is dead. Decide whether to restore or remove both.
Duplicate quality feature	Two separate quality sections exist: <code>/arena_quality</code> (single Wistia video) and <code>/quality</code> (5-tab Vimeo player). Only the second is reachable.
Duplicate gallery images	<code>GalleryPage.tsx</code> : the Interior tab repeats <code>interior7/8/9</code> a second time, and the Elevation tab repeats <code>exteriortwr1-6</code> — visitors see the same six tower renders twice in one carousel. Three Olympus elevation images are imported but commented out of the list (<code>exterior13/14/15</code>).
Slider fragility	<code>PureSlider.tsx:22-41</code> : while showing slide 0 the auto-advance depends entirely on the Wistia JS API binding an <code>"end"</code> event. If Wistia is blocked or slow, no fallback timer is set and the club-house carousel stops permanently on the video. Add a timeout fallback.
Deprecated React prop	<code>VideoSlide.tsx:7</code> passes <code>allowTransparency</code> , which React 19 no longer recognises — logs a console error on <code>/club-house</code> . Remove it (it has had no effect in modern browsers for years).
Bundle splitting	The app ships as one 6.30 MB JS chunk (340 KB gzipped) because all 8 tower datasets — 122,000 lines of coordinate data — are statically imported. Vite emits a chunk-size warning. Route-level <code>React.lazy</code> + dynamic <code>import()</code> per tower would cut initial JS substantially.
Type safety	<code>main.tsx:1-9</code> declares <code>window.global: any</code> and assigns <code>window.global = window</code> — a Node-compatibility shim that is probably no longer needed and weakens the global type. The tower components use <code>floor: any</code> / <code>room: any</code> throughout; the datasets are well-formed and would benefit from a shared exported interface.
README	Still the stock "React + TypeScript + Vite" template text. No setup, deployment or asset-pipeline notes for a project this size.
Toolchain notices	<code>caniuse-lite</code> is 11 months old and <code>baseline-browser-mapping</code> over two months — both warn on every build. Run <code>npx update-browserslist-db@latest</code> .

6. What was verified as working

It is worth recording what testing found in good shape, so effort isn't spent re-checking it:

- **All 8 tower datasets are internally perfect.** 322 floors and 1,747 units across Pavilion, Grandstand, Arcadia, Citadel, Greenfield, Athletica, Stadia and Olympus: no duplicate floor or unit IDs, no unit missing its default / 2D / 2D-static image, no room missing a name or dimension, no empty floors.
- **Every asset import resolves.** All 587 relative import specifiers were checked against the filesystem — zero broken paths, and no `<img>` failed to load in any of the 58 rendered routes.
- **All external media is live** — three Pano2VR tours on `futeservices.com`, six Wistia videos and two Vimeo videos were each loaded in a real browser and confirmed playing.
- **Unit and floor pages handle bad input gracefully** — nonexistent IDs show a clean "Unit not found" / "Floor data not found" screen with a working Go Back button, and unknown paths show the 404 page.
- **Mobile layout holds up.** At 390×844 none of the six pages tested produced horizontal overflow.
- **Initial load of the entrance page is fast** — 0.67 MB, first contentful paint at 528 ms.
- **TypeScript compiles with zero errors** and the production build succeeds in 22.6 s.

7. Recommended order of work

Step	Action	Effort
1	Fix the gallery previous-button index wrap (P0-1) — one line	5 min
2	Move <code>useEffect</code> above the early return in <code>Unit_Pavilion</code> and <code>Unit_Olympus</code> (P0-2)	15 min
3	Restore or re-point the two <code>pano/</code> iframes (P1-1)	30 min – 1 h
4	Add <code>/arena_gallery</code> + <code>/project_status</code> (and the two Golden routes) to the navbar lists; fix <code>lg:bloack</code> ; fix the play/pause icon; remove the 7 <code>console.log</code> calls (P2-3, P2-4, P2-5, P2-7)	1 h
5	Image pipeline: re-encode every asset over ~1.5 MB to WebP capped at 2560 px, and generate real gallery thumbnails (P1-2). Biggest single win for the visitor experience.	1 day
6	De-duplicate the React keys in the three datasets; add the missing Olympus tooltip key (P2-1, P2-2)	1 h
7	Resolve the install conflict, run <code>npm audit fix</code> , update dependencies (P1-3, P1-4)	2 h
8	Standardise unit opening to single click; add <code>alt</code> text and meta/OG tags (P2-6, P2-9)	3 h
9	Delete unused assets and dead files; clear the remaining ESLint errors; add route-level code splitting (P2-10, P3)	1 day

How this was tested. Static analysis: custom import-resolution scanner over 587 specifiers, ESLint, `tsc -b`, unreferenced-asset scan, dataset integrity validator run through Vite's module loader. Runtime: headless Chrome (Puppeteer) against both the Vite dev server and the production `vite preview` build — 58 routes navigated, console errors and failed network requests captured per route, DOM assertions on render output, iframe contents read cross-frame, interaction scripts for the gallery controls and SPA route transitions, mobile-viewport pass at 390×844, and per-route transfer-size measurement with the cache disabled. Network: reachability checks on all 11 external embed and outbound-link URLs.

Note: the Claude-in-Chrome extension could not reach the local dev server in this environment (Chrome returned a network error page for

localhost , 127.0.0.1 and the LAN address alike, while curl received HTTP 200), so all browser testing was performed through a headless Chrome instance driven directly. This is an environment limitation, not a defect in the application.